



BIOLOGY 3

ACADEMIC

Grade 11/12

Curriculum Guide

BIOLOGY 3
Pre-requisites: Biology 1, Biology 2

Course Description:

This course examines the manipulation of genetic information and the evolutionary processes that shape the diversity of life on Earth. Learners explore the principles and techniques of recombinant DNA technology and investigate its applications in medicine, agriculture, and environmental conservation. Through these topics, learners understand how advances in biotechnology contribute to human health, food production, and environmental sustainability.

The course also focuses on the mechanisms and evidence of biological evolution, including genetic variation, natural selection, and speciation. Learners analyze the history of life on Earth using the geologic time scale, examine different lines of evidence supporting evolution, and explain how evolutionary processes lead to the emergence of new species and biodiversity. By the end of the course, learners develop critical thinking and scientific inquiry skills to analyze evolutionary relationships and evaluate the impact of biotechnology on society and the environment.

Elective: Academic

Time Allotment: 80 hours for one term

CONTENT	CONTENT STANDARDS <i>The learners demonstrate an understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
Molecular Genetics and Inheritance		
1. Biological Macromolecules	1. biological macromolecules such as carbohydrates, lipids, proteins, and nucleic acids are essential for cellular functions and metabolism;	1. describe the structure and biological functions of major biomolecules (<i>carbohydrates, lipids, proteins, and nucleic acids</i>) and explain their roles in metabolism and cellular processes;
2. Structure of Genetic Molecules	2. DNA and RNA are nucleic acids that store and transmit genetic information and that their	2. illustrate the molecular structures of DNA, RNA, and proteins and explain how their structures relate to their biological functions;

CONTENT	CONTENT STANDARDS <i>The learners demonstrate an understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
	molecular structures determine their biological functions;	
3. Gene Expression	3. genetic information stored in DNA is expressed through RNA and proteins during the processes of DNA replication, transcription, and translation;	3. explain the flow of genetic information from DNA to RNA to proteins through gene expression;
		4. use diagrams or models to explain the processes of DNA replication, transcription, and translation;
4. Mutations and Genetic Disorders	4. mutations are changes in genetic material that can alter gene expression and protein structure, resulting in variation and genetic disorders;	5. explain how different types of DNA mutations (e.g., substitution, insertion, deletion, and frameshift) affect gene expression and protein function;
		6. use information from secondary sources to investigate genetic disorders caused by mutations or abnormalities in gene expression;
5. Mendelian and Non-Mendelian Inheritance	5. traits are inherited through different genetic patterns that determine the genotype and phenotype of organisms; and	7. use Punnett squares and probability principles to predict genotypes and phenotypes in monohybrid and dihybrid crosses based on Mendelian inheritance;
		8. explain patterns of non-Mendelian inheritance, including incomplete dominance, codominance, multiple alleles, sex-linked, and sex-influenced traits; and
6. Chromosomal Aberrations	6. changes in chromosome structure or number can influence inheritance and may lead to genetic variation or genetic disorders.	9. explain different types of chromosomal aberrations and their effects on inheritance, genetic variation, and human health.
Performance Standards <i>The learners can explain how genetic information is stored, expressed, and inherited by describing the roles of biomolecules and genetic structures, constructing models of gene expression, solving genetic cross problems to predict inheritance patterns, and presenting explanations of how mutations and chromosomal changes affect organism traits and genetic disorders.</i>		
Suggested Performance Tasks		

CONTENT	CONTENT STANDARDS <i>The learners demonstrate an understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
- Conduct a research-based investigation of a genetic disorder using OMIM database (e.g., sickle cell anemia, cystic fibrosis, hemophilia, or Down syndrome). The learners explain the genetic cause of the disorder, the mutation or chromosomal change involved, patterns of inheritance, and its effects on human health through a report, infographic, or presentation. - Construct a physical or digital model illustrating the flow of genetic information from DNA to RNA to proteins through the processes of DNA replication, transcription, and translation, and explain how genetic information leads to the formation of specific traits		
Evolution and Biotechnology		
7. Recombinant DNA Technology	7. recombinant DNA technology allows the modification of genetic material and has applications in medicine, agriculture, and environmental conservation;	10. describe the basic processes involved in recombinant DNA technology; 11. use information from secondary sources to explain applications of recombinant DNA technology in: a. medicine (e.g., recombinant insulin and vaccines); b. agriculture (e.g., genetically modified crops such as Bt corn and Golden Rice); and c. environmental conservation (e.g., bioremediation and oil spill cleanup);
8. Sources of Genetic Variation	8. genetic variation caused by mutation and recombination can lead to evolutionary change in populations;	12. explain how mutations and genetic recombination generate genetic variation that may lead to evolutionary change in populations;
9. History of Life on Earth	9. evolution is supported by scientific evidence and is reflected in the history of life on Earth through the geologic time scale;	13. describe key events in Earth's history using the geologic time scale and identify the major groups of organisms that appeared during these periods;
10. Development of Evolutionary Thought		14. use information from secondary sources to trace the development of evolutionary thought from early ideas to modern evolutionary theory; 15. compare different perspectives about the origin of life, including creation-based views and scientific explanations, and explain how scientific thought is applied to these perspectives;

CONTENT	CONTENT STANDARDS <i>The learners demonstrate an understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
11. Mechanisms and Evidence of Evolution	10. evolutionary processes such as natural selection, genetic drift, mutation, recombination, and speciation explain the emergence of species and the diversity of life on Earth;	16. explain the mechanisms that produce evolutionary change in populations across generations, including natural selection, genetic drift, migration, and mutation;
12. Species and Speciation		17. analyze different lines of evidence for evolution (e.g., fossil record, biogeography, comparative anatomy, embryology, and DNA/protein sequences) to infer evolutionary relationships among organisms;
		18. describe organisms based on biological species concept and reproductive isolation as one of many species concepts;
		19. differentiate the major modes of speciation, including allopatric, sympatric, and parapatric speciation; and
		20. explain how evolutionary processes, including natural selection and speciation, have shaped the diversity of organisms present today.
Performance Standards <i>The learners can analyze evidence of evolution and explain how genetic variation and evolutionary processes lead to speciation and biodiversity. They can also communicate the applications of recombinant DNA technology in medicine, agriculture, and environmental conservation through models, reports, or multimedia presentations.</i>		
Suggested Performance Tasks • Present findings through a poster, infographic, report, or multimedia presentation explaining the scientific basis, applications, and societal benefits of recombinant DNA technology in medicine, agriculture, or environmental conservation (e.g., recombinant insulin, genetically modified crops, or bioremediation). • Construct an evolutionary diagram or phylogenetic tree using evidence such as fossils, comparative anatomy, biogeography, or DNA sequences to explain evolutionary relationships among organisms. • Analyze a case of speciation or biodiversity and present how evolutionary mechanisms such as mutation, natural selection, genetic drift, and reproductive isolation contributed to the emergence of species.		